puzzle has been cleverly labeled the *problem of time* in quantum gravity, and it is where the possibility of emergent time might come to the rescue. If the quantum state of the universe obeys the Wheeler-DeWitt equation (which is plausible, but far from certain), time has to be emergent rather than fundamental.

One way that might work was suggested by Don Page and William Wootters in 1983. Imagine a quantum system consisting of two parts: a clock, and everything else in the universe. Imagine that both the clock and the rest of the system evolve in time as usual. Now take snapshots of the quantum state at regular intervals, perhaps once per second or once per Planck time. In any particular snapshot, the quantum state describes the clock reading some particular time, and the rest of the system in whatever configuration it was in at that time. That gives us a collection of instantaneous quantum states of the system.

The great thing about quantum states is that we can simply add them together (superposing them) to make a new state. So let's make a new quantum state by adding together all of our snapshots. This new quantum state doesn't evolve over time; it just exists, as we constructed it by hand. And there is no specific time reading on the clock; the clock subsystem is in a superposition of all the times at which we took snapshots. It doesn't sound much like our world.

But here's the thing: within that superposition of all the snapshots, the state of the clock is entangled with the state of the rest of the system. If we measure the clock and see that it reads some particular time, then the rest of the universe is in whatever state our original evolving system was caught in at precisely that time.

$$\begin{aligned}\Psi = &(\text{system @ t=0, clock} = 0)\\ &+ (\text{system @ t=1, clock} = 1)\\ &+ (\text{system @ t=2, clock} = 2)\\ &+ \ldots\end{aligned}$$

In other words, there's not "really" time in the superposition state, which is completely static. But entanglement generates a relationship between what the clock reads and what the rest of the universe is doing. And the state of the rest of the universe is precisely what it would be if it were evolving as the original state did over time. We have replaced "time"